

Visualizing and Summarizing a Single Categorical Variable

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Learn how to visualize distributions of categorical variables and we will see how to summarize them using frequency tables.

Load the packages we may need: `tidyverse`, `ggdist`, and `ggribes`. They should already be installed for you, but if not, go to the console below this window and type `install.packages("tidyverse")`, etc. You can also install multiple packages at once using the `c` function to make a vector like this: `install.packages(c("tidyverse", "ggdist", "ggribes"))`

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.4
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()      masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggdist)
```

```
library(ggribes)
```

```
##
```

```
## Attaching package: 'ggribes'
```

```
##
```

```
## The following objects are masked from 'package:ggdist':
```

```
##
```

```
##   scale_point_color_continuous, scale_point_color_discrete,
```

```
##   scale_point_colour_continuous, scale_point_colour_discrete,
```

```
##   scale_point_fill_continuous, scale_point_fill_discrete,
```

```
##   scale_point_size_continuous
```

Today, we will work with the `GSS_clean` dataset that contains data from the General Social Survey, which is a nationally representative survey of Americans to “monitor and explain trends in opinions, attitudes, and behaviors”.

I have put a file called `GSS_clean` into the `data` folder in this project. Load the dataset here, and call it `GSS`:

```
GSS <- read.csv("/cloud/project/data/GSS_clean.csv", header = TRUE)
```

```
head(GSS)
```

```
##   respondent_id_number number_of_children age_of_respondent
## 1                    1                0                43
## 2                    2                3                74
## 3                    3                2                42
## 4                    4                2                63
```

##	5	5	0	71
##	6	6	2	67
##	highest_year_of_school_completed highest_year_school_completed_father			
##	1	14		12
##	2	10		0
##	3	16		12
##	4	16		16
##	5	18		12
##	6	16		12
##	highest_year_school_completed_mother highest_year_school_completed_spouse			
##	1	12		NA
##	2	8		NA
##	3	16		14
##	4	12		14
##	5	12		NA
##	6	12		NA
##	college_major_1	college_major_2	diploma_ged_or_other	respondents_sex
##	1	Communications/speech	<NA> High school diploma	Male
##	2	<NA>	<NA> Ged	Female
##	3	Electronics	<NA> High school diploma	Male
##	4	Business administration	Computer science High school diploma	Female
##	5	Business administration	<NA> High school diploma	Male
##	6	Mathmatics	<NA> High school diploma	Female
##	number_of_brothers_and_sisters labor_force_status			
##	1	4	Temp not working	
##	2	4	Retired	
##	3	2	Working fulltime	
##	4	3	Working fulltime	
##	5	3	Retired	
##	6	1	Retired	
##	number_of_hours_worked_last_week number_of_hours_usually_work_a_week			
##	1	NA		41
##	2	NA		NA
##	3	40		NA
##	4	40		NA
##	5	NA		NA
##	6	NA		NA
##	self_emp_or_works_for_somebody govt_or_private_employee			
##	1	Someone else	Private	
##	2	Someone else	Private	
##	3	Someone else	Private	
##	4	Someone else	Private	
##	5	Someone else	Private	
##	6	Someone else	Private	
##	occupation_code marital_status			
##	1	Human resources workers	Never married	
##	2	Packers and packagers, hand	Separated	
##	3	Computer network architects	Married	
##	4	Diagnostic related technologists and technicians	Married	
##	5	Human resources managers	Divorced	
##	6	Financial managers	Widowed	
##	marital_type race_of_respondent born_in_us			
##	1	<NA>	White	Yes
##	2	<NA>	White	Yes

## 3	Marriage between a man and a woman	White	Yes
## 4	Marriage between a man and a woman	White	Yes
## 5	<NA>	Black	Yes
## 6	<NA>	White	Yes
##	should_govt_reduce_income_differences		
## 1	<NA>		
## 2	No govt action		
## 3	3		
## 4	5		
## 5	6		
## 6	<NA>		
##	favor_or_oppose_death_penalty_for_murder	should_marijuana_be_made_legal	
## 1	Favor	<NA>	
## 2	Oppose	Not legal	
## 3	Favor	Legal	
## 4	Oppose	Not legal	
## 5	Oppose	Not legal	
## 6	Favor	<NA>	
##	rs_religious_preference	specific_denomination	other_protestant_denominations
## 1	Christian	No denomination	<NA>
## 2	Catholic	<NA>	<NA>
## 3	None	<NA>	<NA>
## 4	Protestant	United methodist	<NA>
## 5	Catholic	<NA>	<NA>
## 6	Catholic	<NA>	<NA>
##	general_happiness	condition_of_health	is_life_exciting_or_dull
## 1	Pretty happy	Good	Routine
## 2	Very happy	Excellent	Exciting
## 3	Very happy	<NA>	<NA>
## 4	Very happy	<NA>	<NA>
## 5	Pretty happy	Excellent	Routine
## 6	Not too happy	Good	Routine
##	ballot_used_for_interview	parks_and_recreation	mass_transportation
## 1	Ballot a	Too much	About right
## 2	Ballot c	About right	About right
## 3	Ballot b	About right	Too little
## 4	Ballot b	About right	About right
## 5	Ballot c	Too little	Too little
## 6	Ballot a	About right	Too little
##	number_of_persons_in_household	total_family_income	respondents_income
## 1	5	Refused	Refused
## 2	2	\$25000 or more	(Missing)
## 3	4	\$25000 or more	\$25000 or more
## 4	2	Refused	\$25000 or more
## 5	2	Refused	(Missing)
## 6	2	(Missing)	(Missing)
##	region_of_interview	size_of_place_in_1000s	political_party_affiliation
## 1	New England	14	Republican
## 2	New England	14	Independent
## 3	New England	14	Independent
## 4	New England	14	Independent
## 5	New England	14	<NA>
## 6	New England	14	Independent
##	think_of_self_as_liberal_or_conservative	space_exploration_program	

```
## 1          Conservative          Too much
## 2          (Missing)          <NA>
## 3      Slightly conservative      Too little
## 4          Moderate          <NA>
## 5      Extrmly conservative      <NA>
## 6      Slightly liberal          <NA>
##  improving_protecting_environment improving_protecting_nations_health
## 1          About right          Too little
## 2          <NA>          <NA>
## 3          Too little          Too little
## 4          <NA>          <NA>
## 5          <NA>          <NA>
## 6          <NA>          <NA>
##  lived_in_us
## 1          NA
## 2          NA
## 3          NA
## 4          NA
## 5          NA
## 6          NA
```

Explore the dataset in the usual ways.

```
head(GSS, 10)
```

```
##  respondent_id_number number_of_children age_of_respondent
## 1          1          0          43
## 2          2          3          74
## 3          3          2          42
## 4          4          2          63
## 5          5          0          71
## 6          6          2          67
## 7          7          6          59
## 8          8          0          43
## 9          9          4          62
## 10         10         2          55
##  highest_year_of_school_completed highest_year_school_completed_father
## 1          14          12
## 2          10          0
## 3          16          12
## 4          16          16
## 5          18          12
## 6          16          12
## 7          13          NA
## 8          12          NA
## 9          8          12
## 10         12          6
##  highest_year_school_completed_mother highest_year_school_completed_spouse
## 1          12          NA
## 2          8          NA
## 3          16          14
## 4          12          14
## 5          12          NA
## 6          12          NA
## 7          6          NA
```

## 8		12	NA
## 9		12	NA
## 10		20	11
##	college_major_1	college_major_2	diploma_ged_or_other
## 1	Communications/speech	<NA>	High school diploma
## 2	<NA>	<NA>	Ged
## 3	Electronics	<NA>	High school diploma
## 4	Business administration	Computer science	High school diploma
## 5	Business administration	<NA>	High school diploma
## 6	Mathmatics	<NA>	High school diploma
## 7	<NA>	<NA>	High school diploma
## 8	<NA>	<NA>	High school diploma
## 9	<NA>	<NA>	<NA>
## 10	<NA>	<NA>	High school diploma
##	respondents_sex	number_of_brothers_and_sisters	labor_force_status
## 1	Male		4 Temp not working
## 2	Female		4 Retired
## 3	Male		2 Working fulltime
## 4	Female		3 Working fulltime
## 5	Male		3 Retired
## 6	Female		1 Retired
## 7	Female		7 Working fulltime
## 8	Male		2 Working fulltime
## 9	Female		4 Working fulltime
## 10	Male		2 Working fulltime
##	number_of_hours_worked_last_week	number_of_hours_usually_work_a_week	
## 1		NA	41
## 2		NA	NA
## 3		40	NA
## 4		40	NA
## 5		NA	NA
## 6		NA	NA
## 7		35	NA
## 8		89	NA
## 9		40	NA
## 10		40	NA
##	self_emp_or_works_for_somebody	govt_or_private_employee	
## 1	Someone else	Private	
## 2	Someone else	Private	
## 3	Someone else	Private	
## 4	Someone else	Private	
## 5	Someone else	Private	
## 6	Someone else	Private	
## 7	Someone else	Government	
## 8	Someone else	Private	
## 9	Self-employed	Private	
## 10	Someone else	Private	
##		occupation_code	marital_status
## 1		Human resources workers	Never married
## 2		Packers and packagers, hand	Separated
## 3		Computer network architects	Married
## 4	Diagnostic related technologists and technicians		Married
## 5		Human resources managers	Divorced
## 6		Financial managers	Widowed

## 7	Nursing, psychiatric, and home health aides	Divorced
## 8	Shipping, receiving, and traffic clerks	Never married
## 9	Childcare workers	Widowed
## 10	Elevator installers and repairers	Married
##	marital_type race_of_respondent born_in_us	
## 1	<NA>	White Yes
## 2	<NA>	White Yes
## 3	Marriage between a man and a woman	White Yes
## 4	Marriage between a man and a woman	White Yes
## 5	<NA>	Black Yes
## 6	<NA>	White Yes
## 7	<NA>	Black Yes
## 8	<NA>	White Yes
## 9	<NA>	White Yes
## 10	Marriage between a man and a woman	White Yes
##	should_govt_reduce_income_differences	
## 1	<NA>	
## 2	No govt action	
## 3	3	
## 4	5	
## 5	6	
## 6	<NA>	
## 7	5	
## 8	Govt reduce diff	
## 9	3	
## 10	3	
##	favor_or_oppose_death_penalty_for_murder should_marijuana_be_made_legal	
## 1	Favor	<NA>
## 2	Oppose	Not legal
## 3	Favor	Legal
## 4	Oppose	Not legal
## 5	Oppose	Not legal
## 6	Favor	<NA>
## 7	Favor	Legal
## 8	Favor	Legal
## 9	Favor	Legal
## 10	Favor	Legal
##	rs_religious_preference specific_denomination other_protestant_denominations	
## 1	Christian	No denomination <NA>
## 2	Catholic	<NA> <NA>
## 3	None	<NA> <NA>
## 4	Protestant	United methodist <NA>
## 5	Catholic	<NA> <NA>
## 6	Catholic	<NA> <NA>
## 7	Protestant	Baptist-dk which <NA>
## 8	Catholic	<NA> <NA>
## 9	None	<NA> <NA>
## 10	Protestant	Other Congregationalist, 1st Congreg
##	general_happiness condition_of_health is_life_exciting_or_dull	
## 1	Pretty happy	Good Routine
## 2	Very happy	Excellent Exciting
## 3	Very happy	<NA> <NA>
## 4	Very happy	<NA> <NA>
## 5	Pretty happy	Excellent Routine

## 6	Not too happy	Good	Routine
## 7	Pretty happy	Good	Routine
## 8	Pretty happy	Excellent	Dull
## 9	Not too happy	<NA>	<NA>
## 10	Pretty happy	<NA>	<NA>
##	ballot_used_for_interview	parks_and_recreation	mass_transportation
## 1	Ballot a	Too much	About right
## 2	Ballot c	About right	About right
## 3	Ballot b	About right	Too little
## 4	Ballot b	About right	About right
## 5	Ballot c	Too little	Too little
## 6	Ballot a	About right	Too little
## 7	Ballot c	Too little	About right
## 8	Ballot c	Too little	About right
## 9	Ballot b	Too little	Too little
## 10	Ballot b	About right	About right
##	number_of_persons_in_household	total_family_income	respondents_income
## 1		5	Refused
## 2		2	\$25000 or more
## 3		4	\$25000 or more
## 4		2	Refused
## 5		2	Refused
## 6		2	(Missing)
## 7		2	\$15000 - 19999
## 8		1	\$25000 or more
## 9		1	\$5000 to 5999
## 10		2	\$25000 or more
##	region_of_interview	size_of_place_in_1000s	political_party_affiliation
## 1	New England	14	Republican
## 2	New England	14	Independent
## 3	New England	14	Independent
## 4	New England	14	Independent
## 5	New England	14	<NA>
## 6	New England	14	Independent
## 7	Middle Atlantic	24	Democrat
## 8	Middle Atlantic	24	Republican
## 9	Middle Atlantic	24	<NA>
## 10	New England	56	Democrat
##	think_of_self_as_liberal_or_conservative	space_exploration_program	
## 1	Conservative	Too much	
## 2	(Missing)	<NA>	
## 3	Slightly conservative	Too little	
## 4	Moderate	<NA>	
## 5	Extrmly conservative	<NA>	
## 6	Slightly liberal	<NA>	
## 7	Moderate	<NA>	
## 8	Slightly conservative	Too little	
## 9	Moderate	Too much	
## 10	(Missing)	<NA>	
##	improving_protecting_environment	improving_protecting_nations_health	
## 1	About right	Too little	
## 2	<NA>	<NA>	
## 3	Too little	Too little	
## 4	<NA>	<NA>	

```
## 5          <NA>          <NA>
## 6          <NA>          <NA>
## 7          <NA>          <NA>
## 8          Too little    Too little
## 9          About right   Too little
## 10         <NA>         <NA>
## lived_in_us
## 1          NA
## 2          NA
## 3          NA
## 4          NA
## 5          NA
## 6          NA
## 7          NA
## 8          NA
## 9          NA
## 10         NA
```

```
tail(GSS, 10)
```

```
##      respondent_id_number number_of_children age_of_respondent
## 2339          2339          3          46
## 2340          2340          0          29
## 2341          2341          2      89 or older
## 2342          2342          2          68
## 2343          2343          0          19
## 2344          2344          2          37
## 2345          2345          4          75
## 2346          2346          3          67
## 2347          2347          2          72
## 2348          2348          6          79
##      highest_year_of_school_completed highest_year_school_completed_father
## 2339          12          12
## 2340          14          10
## 2341           8           6
## 2342          14          12
## 2343          12          12
## 2344          12          12
## 2345          12           8
## 2346          12          12
## 2347          16          14
## 2348          12           8
##      highest_year_school_completed_mother highest_year_school_completed_spouse
## 2339          NA          NA
## 2340          16          10
## 2341          NA          NA
## 2342          14          NA
## 2343          16          NA
## 2344          13          NA
## 2345           8           8
## 2346          14          12
## 2347          12          14
## 2348           8          NA
##      college_major_1 college_major_2 diploma_ged_or_other respondents_sex
## 2339          <NA>          <NA> High school diploma      Female
```


##	2340	<NA>	<NA>	High school diploma	Female
##	2341	<NA>	<NA>	<NA>	Male
##	2342	<NA>	<NA>	High school diploma	Female
##	2343	<NA>	<NA>	High school diploma	Male
##	2344	<NA>	<NA>	High school diploma	Female
##	2345	<NA>	<NA>	High school diploma	Female
##	2346	<NA>	<NA>	High school diploma	Female
##	2347	<NA>	<NA>	High school diploma	Male
##	2348	<NA>	<NA>	High school diploma	Female
##		number_of_brothers_and_sisters	labor_force_status		
##	2339		2	Keeping house	
##	2340		1	Working fulltime	
##	2341		6	Retired	
##	2342		4	Retired	
##	2343		3	School	
##	2344		9	Working fulltime	
##	2345		9	Working parttime	
##	2346		3	Retired	
##	2347		1	Retired	
##	2348		10	Keeping house	
##		number_of_hours_worked_last_week	number_of_hours_usually_work_a_week		
##	2339		NA		NA
##	2340		60		NA
##	2341		NA		NA
##	2342		NA		NA
##	2343		NA		NA
##	2344		36		NA
##	2345		36		NA
##	2346		NA		NA
##	2347		NA		NA
##	2348		NA		NA
##		self_emp_or_works_for_somebody	govt_or_private_employee		
##	2339	Someone else	Government		
##	2340	Someone else	Private		
##	2341	Someone else	Private		
##	2342	Someone else	Private		
##	2343	<NA>	<NA>		
##	2344	Someone else	Government		
##	2345	Someone else	Private		
##	2346	Someone else	Private		
##	2347	Someone else	Private		
##	2348	Someone else	Government		
##			occupation_code		
##	2339		Nursing, psychiatric, and home health aides		
##	2340		Retail salespersons		
##	2341		Driver/sales workers and truck drivers		
##	2342		First-line supervisors of construction trades and extraction workers		
##	2343		<NA>		
##	2344		Secretaries and administrative assistants		
##	2345		Cashiers		
##	2346		Payroll and timekeeping clerks		
##	2347		Drafters		
##	2348		Cooks		
##		marital_status	marital_type	race_of_respondent	

##	2339	Divorced	<NA>	White
##	2340	Married Marriage between a man and a woman		White
##	2341	Widowed	<NA>	White
##	2342	Divorced	<NA>	White
##	2343	Never married	<NA>	White
##	2344	Divorced	<NA>	White
##	2345	Married Marriage between a man and a woman		White
##	2346	Married Marriage between a man and a woman		White
##	2347	Married Marriage between a man and a woman		White
##	2348	Widowed	<NA>	White
##	born_in_us should_govt_reduce_income_differences			
##	2339	Yes	Govt reduce diff	
##	2340	Yes	<NA>	
##	2341	Yes	Govt reduce diff	
##	2342	Yes	<NA>	
##	2343	Yes	4	
##	2344	Yes	<NA>	
##	2345	Yes	<NA>	
##	2346	Yes	4	
##	2347	Yes	No govt action	
##	2348	Yes	<NA>	
##	favor_or_oppose_death_penalty_for_murder should_marijuana_be_made_legal			
##	2339		Favor	Legal
##	2340		Favor	<NA>
##	2341		Favor	Not legal
##	2342		Oppose	<NA>
##	2343		Favor	Legal
##	2344		Oppose	<NA>
##	2345		Favor	<NA>
##	2346		Favor	Legal
##	2347		Favor	Not legal
##	2348		Favor	<NA>
##	rs_religious_preference specific_denomination			
##	2339	Protestant	Am bapt ch in usa	
##	2340	Other	<NA>	
##	2341	Protestant	<NA>	
##	2342	None	<NA>	
##	2343	None	<NA>	
##	2344	Protestant	<NA>	
##	2345	Protestant	Other methodist	
##	2346	Jewish	<NA>	
##	2347	Protestant	<NA>	
##	2348	Catholic	<NA>	
##	other_protestant_denominations general_happiness condition_of_health			
##	2339	<NA>	Not too happy	Fair
##	2340	<NA>	Pretty happy	Fair
##	2341	<NA>	Pretty happy	<NA>
##	2342	<NA>	Pretty happy	Fair
##	2343	<NA>	Pretty happy	Good
##	2344	<NA>	Pretty happy	Good
##	2345	<NA>	Pretty happy	Excellent
##	2346	<NA>	Very happy	Excellent
##	2347	<NA>	Very happy	<NA>
##	2348	<NA>	Very happy	Good

##	is_life_exciting_or_dull	ballot_used_for_interview	parcs_and_recreation
## 2339	Routine	Ballot c	Too little
## 2340	Exciting	Ballot a	About right
## 2341	<NA>	Ballot b	Too much
## 2342	Exciting	Ballot a	About right
## 2343	Routine	Ballot c	About right
## 2344	Routine	Ballot a	Too much
## 2345	Routine	Ballot a	About right
## 2346	Routine	Ballot c	About right
## 2347	<NA>	Ballot b	About right
## 2348	Routine	Ballot a	Too little
##	mass_transportation	number_of_persons_in_household	total_family_income
## 2339	Too little	2	\$15000 - 19999
## 2340	<NA>	2	\$25000 or more
## 2341	Too much	1	(Missing)
## 2342	About right	1	\$8000 to 9999
## 2343	About right	4	\$15000 - 19999
## 2344	About right	3	Refused
## 2345	About right	2	\$25000 or more
## 2346	Too little	2	\$25000 or more
## 2347	About right	2	\$25000 or more
## 2348	Too little	1	Refused
##	respondents_income	region_of_interview	size_of_place_in_1000s
## 2339	(Missing)	New England	5
## 2340	\$25000 or more	New England	5
## 2341	(Missing)	New England	5
## 2342	(Missing)	New England	5
## 2343	(Missing)	New England	5
## 2344	Refused	New England	1
## 2345	\$15000 - 19999	New England	1
## 2346	\$3000 to 3999	New England	1
## 2347	(Missing)	New England	1
## 2348	(Missing)	New England	1
##	political_party_affiliation	think_of_self_as_liberal_or_conservative	
## 2339	<NA>		Moderate
## 2340	<NA>		Moderate
## 2341	Independent		Moderate
## 2342	<NA>		(Missing)
## 2343	Republican		Moderate
## 2344	<NA>		Moderate
## 2345	Republican	Slightly conservative	
## 2346	<NA>		Moderate
## 2347	Independent		Conservative
## 2348	<NA>		(Missing)
##	space_exploration_program	improving_protecting_environment	
## 2339	<NA>	Too little	
## 2340	Too little	Too little	
## 2341	<NA>	Too little	
## 2342	<NA>	<NA>	
## 2343	About right	About right	
## 2344	Too much	Too little	
## 2345	Too much	About right	
## 2346	<NA>	<NA>	
## 2347	About right	About right	

```
## 2348                <NA>                <NA>
##      improving_protecting_nations_health lived_in_us
## 2339                Too little            NA
## 2340                About right           NA
## 2341                <NA>                 NA
## 2342                <NA>                 NA
## 2343                About right           NA
## 2344                Too little            NA
## 2345                Too little            NA
## 2346                <NA>                 NA
## 2347                Too little            NA
## 2348                <NA>                 NA
```

summary(GSS)

```
## respondent_id_number number_of_children age_of_respondent
## Min.      : 1.0      Length:2348      Length:2348
## 1st Qu.: 587.8      Class :character  Class :character
## Median :1174.5      Mode  :character  Mode  :character
## Mean    :1174.5
## 3rd Qu.:1761.2
## Max.    :2348.0
##
## highest_year_of_school_completed highest_year_school_completed_father
## Min.      : 0.00      Min.      : 0.00
## 1st Qu.:12.00      1st Qu.:10.00
## Median :14.00      Median :12.00
## Mean    :13.73      Mean    :11.88
## 3rd Qu.:16.00      3rd Qu.:14.00
## Max.    :20.00      Max.    :20.00
## NA's     :3         NA's      :661
## highest_year_school_completed_mother highest_year_school_completed_spouse
## Min.      : 0.00      Min.      : 0.00
## 1st Qu.:11.00      1st Qu.:12.00
## Median :12.00      Median :14.00
## Mean    :11.88      Mean    :13.89
## 3rd Qu.:14.00      3rd Qu.:16.00
## Max.    :20.00      Max.    :20.00
## NA's     :259       NA's      :1360
## college_major_1 college_major_2 diploma_ged_or_other respondents_sex
## Length:2348      Length:2348      Length:2348      Length:2348
## Class :character  Class :character  Class :character  Class :character
## Mode  :character  Mode  :character  Mode  :character  Mode  :character
##
##
##
## number_of_brothers_and_sisters labor_force_status
## Min.      : 0.00      Length:2348
## 1st Qu.: 2.00      Class :character
## Median : 3.00      Mode  :character
## Mean    : 3.58
## 3rd Qu.: 5.00
## Max.    :25.00
## NA's     :5
```

```

## number_of_hours_worked_last_week number_of_hours_usually_work_a_week
## Min. : 1.00 Min. : 6.00
## 1st Qu.:35.00 1st Qu.:35.00
## Median :40.00 Median :40.00
## Mean :41.28 Mean :38.25
## 3rd Qu.:50.00 3rd Qu.:42.75
## Max. :89.00 Max. :84.00
## NA's :967 NA's :2300
## self_emp_or_works_for_somebody govt_or_private_employee occupation_code
## Length:2348 Length:2348 Length:2348
## Class :character Class :character Class :character
## Mode :character Mode :character Mode :character
##
##
##
## marital_status marital_type race_of_respondent born_in_us
## Length:2348 Length:2348 Length:2348 Length:2348
## Class :character Class :character Class :character Class :character
## Mode :character Mode :character Mode :character Mode :character
##
##
##
## should_govt_reduce_income_differences favor_or_oppose_death_penalty_for_murder
## Length:2348 Length:2348
## Class :character Class :character
## Mode :character Mode :character
##
##
##
## should_marijuana_be_made_legal rs_religious_preference specific_denomination
## Length:2348 Length:2348 Length:2348
## Class :character Class :character Class :character
## Mode :character Mode :character Mode :character
##
##
##
## other_protestant_denominations general_happiness condition_of_health
## Length:2348 Length:2348 Length:2348
## Class :character Class :character Class :character
## Mode :character Mode :character Mode :character
##
##
##
## is_life_exciting_or_dull ballot_used_for_interview parks_and_recreation
## Length:2348 Length:2348 Length:2348
## Class :character Class :character Class :character
## Mode :character Mode :character Mode :character
##
##

```

```
##
##
## mass_transportation number_of_persons_in_household total_family_income
## Length:2348      Min.   : 1.000      Length:2348
## Class :character  1st Qu.: 1.000      Class :character
## Mode  :character  Median : 2.000      Mode  :character
##                  Mean   : 2.352
##                  3rd Qu.: 3.000
##                  Max.   :11.000
##
## respondents_income region_of_interview size_of_place_in_1000s
## Length:2348      Length:2348      Min.   : 0.0
## Class :character  Class :character  1st Qu.: 9.0
## Mode  :character  Mode  :character  Median : 30.0
##                  Mean   : 336.9
##                  3rd Qu.: 127.0
##                  Max.   :8175.0
##
## political_party_affiliation think_of_self_as_liberal_or_conservative
## Length:2348      Length:2348
## Class :character  Class :character
## Mode  :character  Mode  :character
##
##
##
## space_exploration_program improving_protecting_environment
## Length:2348      Length:2348
## Class :character  Class :character
## Mode  :character  Mode  :character
##
##
##
## improving_protecting_nations_health lived_in_us
## Length:2348      Min.   : 1.00
## Class :character  1st Qu.:14.00
## Mode  :character  Median :21.00
##                  Mean   :25.49
##                  3rd Qu.:37.00
##                  Max.   :93.00
##                  NA's   :2051
##
```

How many cases are in this dataset? How many variables?

```
n_cases <- nrow(GSS)
n_cases
```

```
## [1] 2348
```

There are 2348 cases, and 45 variables within the dataframe.

Name some variables in this dataset that are categorical?

Marital status, respondent sex, general happiness, ... ### Do any of the variables in this dataset have missing values? How can you tell?

```
# check missing values
any_missing_values <- any(is.na(GSS))
any_missing_values
```

```
## [1] TRUE
```

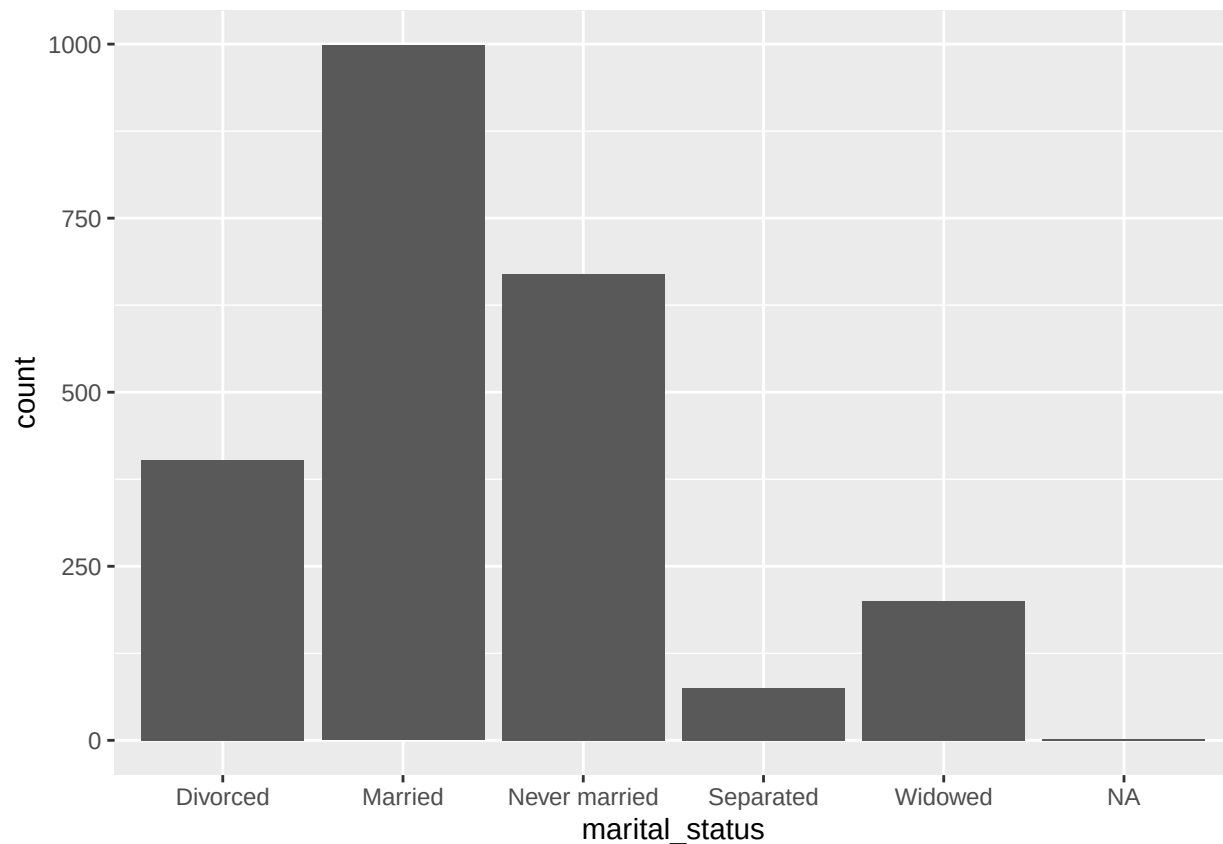
Yes, by running the chunk above it will return the value ‘True’ if there is a NA value present within the dataframe.

Next, we will create a first attempt at a bar graph for the single categorical variable, `marital_status` from this dataset. Then, we will embark on a sequence of tasks that will lead us to create an improved bar graph for this variable.

Complete the code below, filling in the name of the dataframe, setting `x` = the variable we want to examine, and using `geom_bar()`.

Barplot of a single categorical variable

```
ggplot(data = GSS, mapping = aes(x = marital_status)) +
  geom_bar()
```



Notice that one of the levels of the variable `marital_status` is NA. It may be worth exploring the meaning of this level and why survey respondents selected this response over any of the others. However, for our purposes today, we are going to drop this level of `marital_status`.

It turns out that the `general_happiness` variable also has a level called `NA` and we will drop that also, for today. The code for doing this is below. You can adapt this code any time you need to drop a level of a categorical variable. One way to do that is to use an `&` (for the logical “and”) inside the filter function and add a second critereon.

Notice that I am creating a new dataframe called `GSS2` where I am keeping all but the level `NA` for `marital_status`. The `!=` notation means “not equal to”.

```
GSS2<- GSS %>%  
  filter(marital_status != "NA") %>%  
  droplevels()
```

You can check that the variables have been dropped by piping the new dataframe, `GSS2` into the `distinct` function which lists the various levels for a categorical variable. Fill in the variable name `marital_status` below.

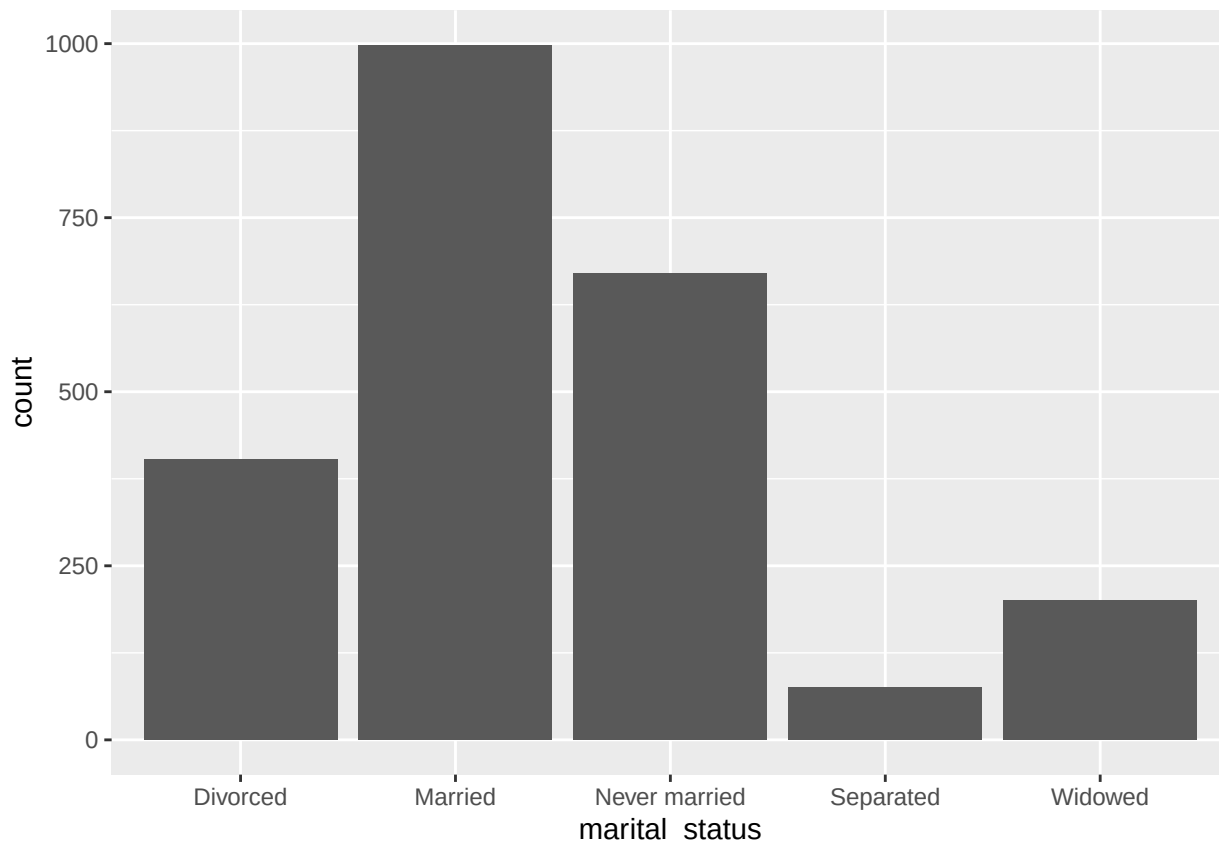
```
GSS2 %>%  
  distinct(marital_status)
```

```
## marital_status  
## 1 Never married  
## 2 Separated  
## 3 Married  
## 4 Divorced  
## 5 Widowed
```

From now on, remember to use the new dataframe `GSS2` instead of the original one.

Below, make an updated version of the bar graph for `marital_status` and verify that the `NA` level is gone. Remember to use `GSS2`!

```
ggplot(data = GSS2, mapping = aes(x = marital_status)) +  
  geom_bar()
```

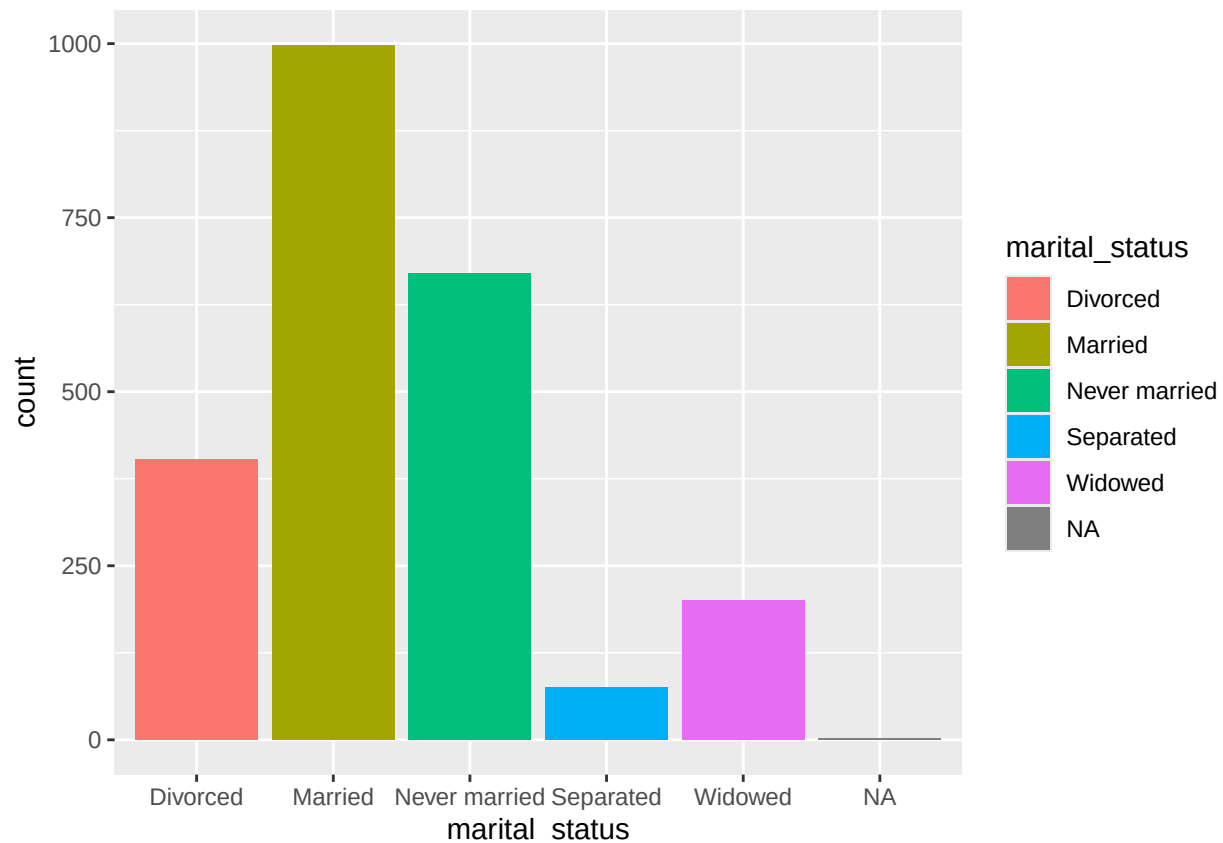
Write a few sentences to describe the distribution of the `marital_status` variable. Start with a who/what/when/where sentence. Next, point out what you think is the most important feature of the graph, supporting what you say with numbers. Then, tell the next most important thing, again, supporting what you say, and so on.

The bar chart above displays the distribution of marital status categories along with the number of respondents. Most of the data is concentrated on the left side of the graph, indicating that the majority of respondents fall into the categories of married, never married, or divorced. These categories roughly fall within the first and second quartiles of the dataset. Meanwhile, fewer individuals are separated or widowed, placing them in the third and fourth quartiles.

Add color to the bars.

We can color the bars by adding the argument `fill = marital_status` inside the aesthetic for the bar graph. Do this below.

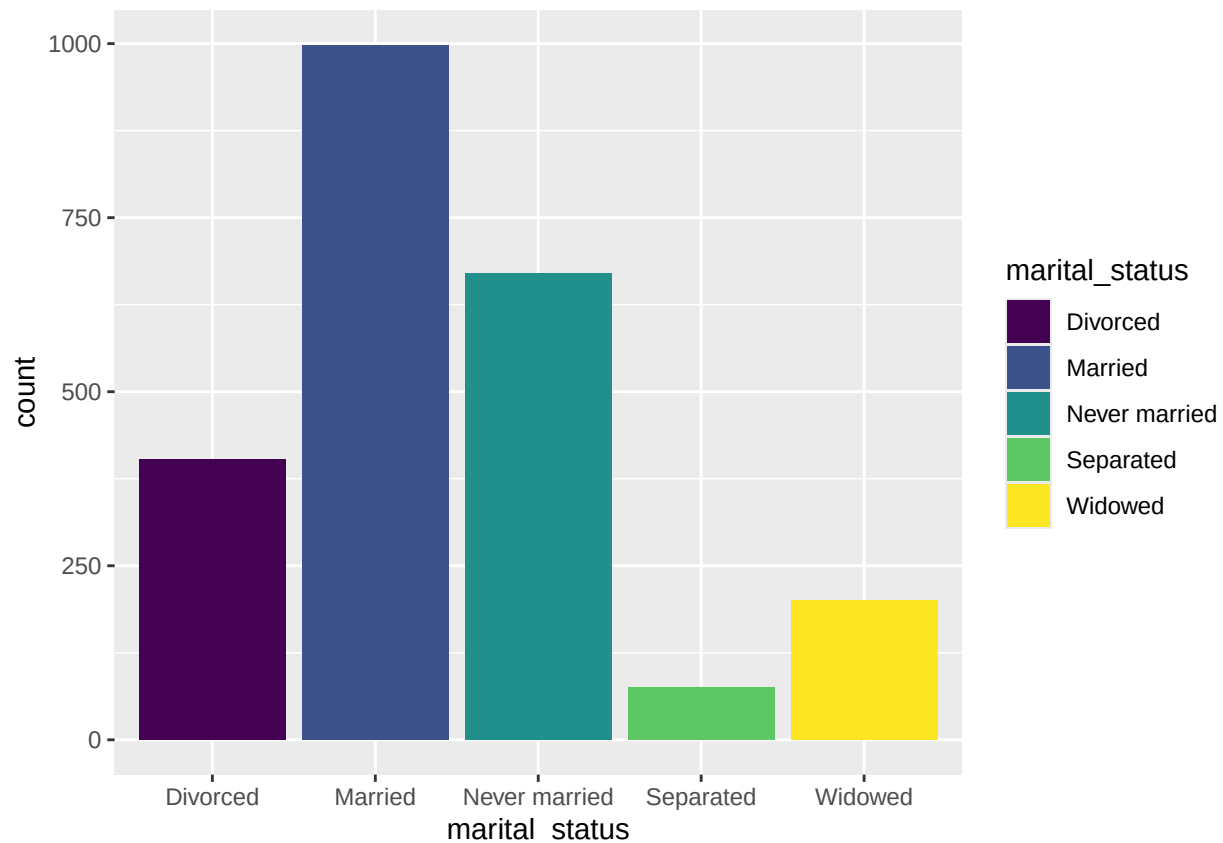
```
ggplot(data = GSS, mapping = aes(x = marital_status, fill = marital_status)) +  
  geom_bar()
```



Next, we can use better colors by adding one more line of code to the end of the code for the bar graph above. The “Viridis” color palates are specially designed for visibility for everyone, but especially for colorblind readers. You can read more about the options for this function, as well as see comparisons of all the different color scales, at <https://cran.r-project.org/web/packages/viridis/vignettes/intro-to-viridis.html>.

The extra line is `scale_fill_viridis_d()`. Remember to put a `+` symbol at the end of the previous line.

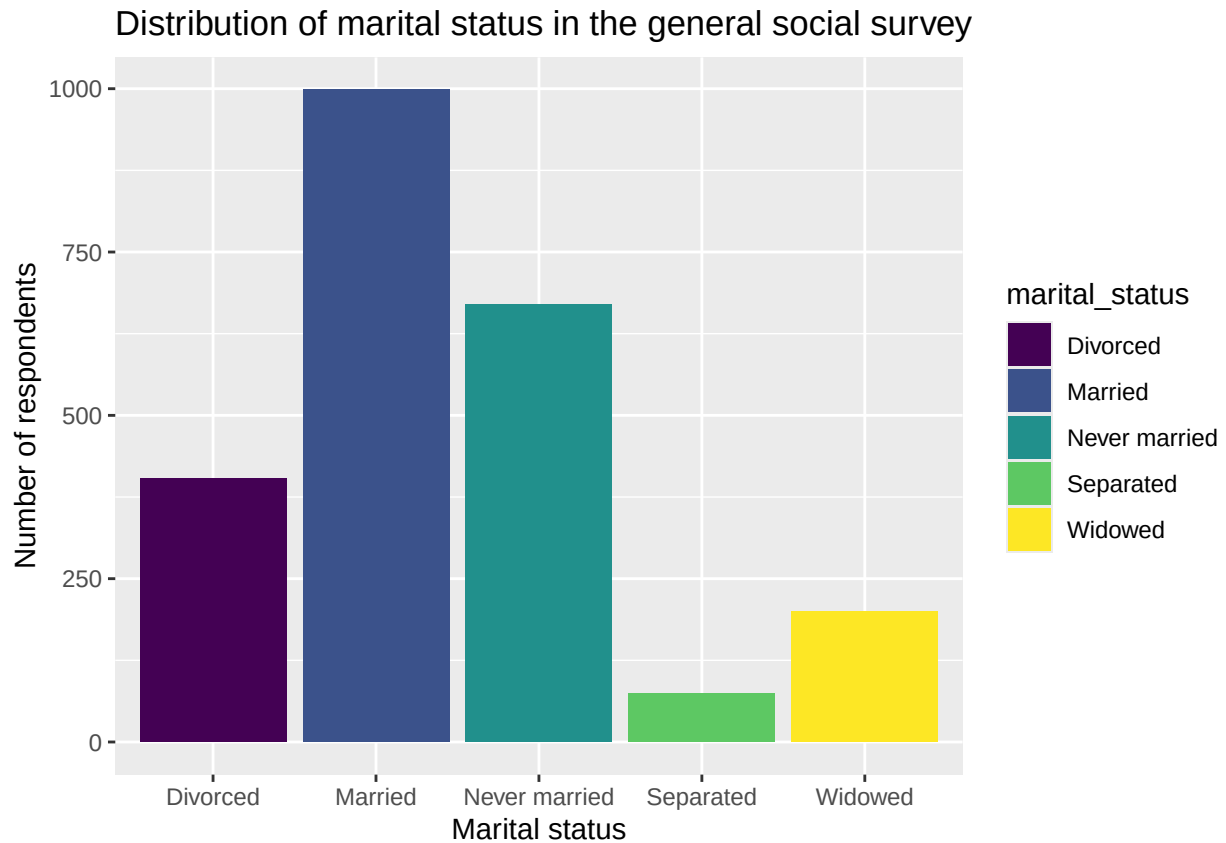
```
ggplot(data = GSS2, mapping = aes(x = marital_status, fill = marital_status)) +  
  geom_bar() +  
  scale_fill_viridis_d()
```



Add labels and a title

Build on what we have just done by adding nice labels and a title to the graph. You will use all of the same code as before, but add another layer using another `+`. The line of code for the new layer should look like: `labs(x = "x axis title goes here", y = "y-axis title goes here", title = "title for the whole graph goes here")`. Add appropriate axis labels and a title to your graph below.

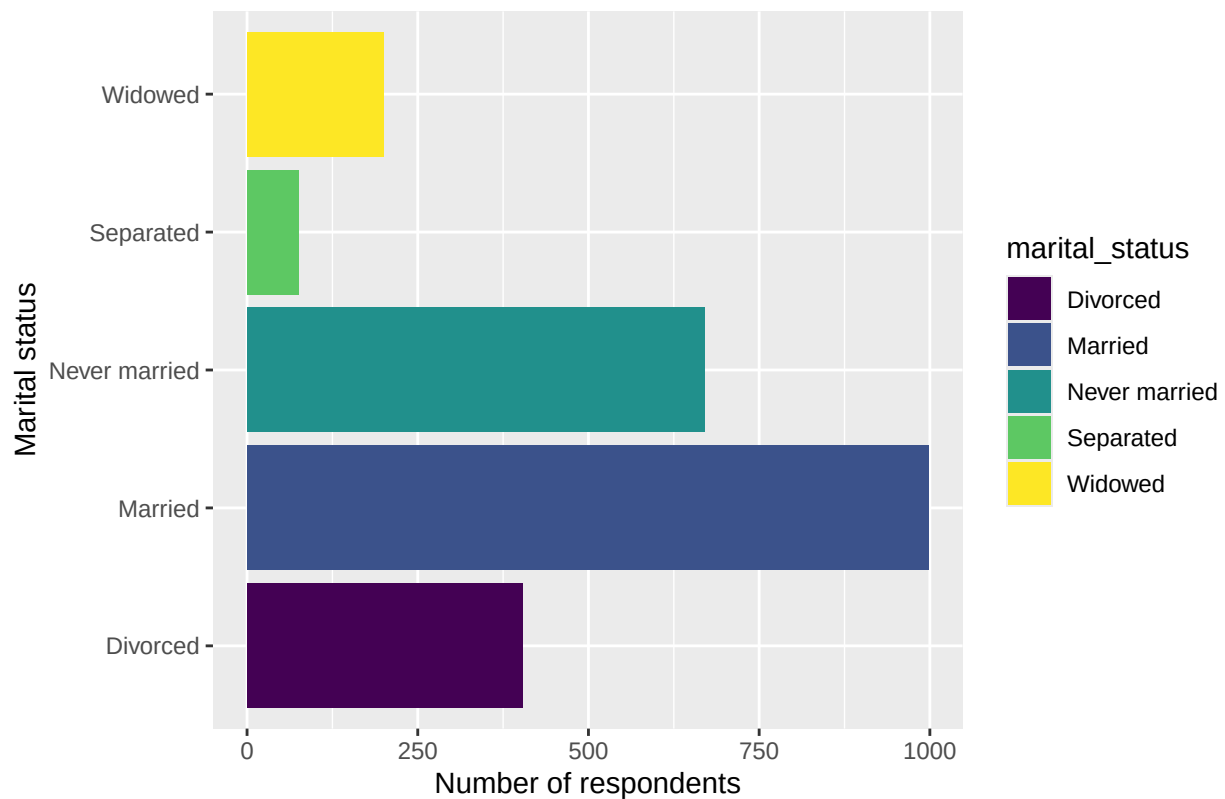
```
ggplot(data = GSS2, mapping = aes(x = marital_status, fill = marital_status)) +  
  geom_bar() +  
  scale_fill_viridis_d() + labs(x = "Marital status", y = "Number of respondents", title = "Distribution of marital status")
```



Exercise: in the space below, create another version of this bar graph, where the bars run horizontally instead of vertically. Make sure to update your axis labels!

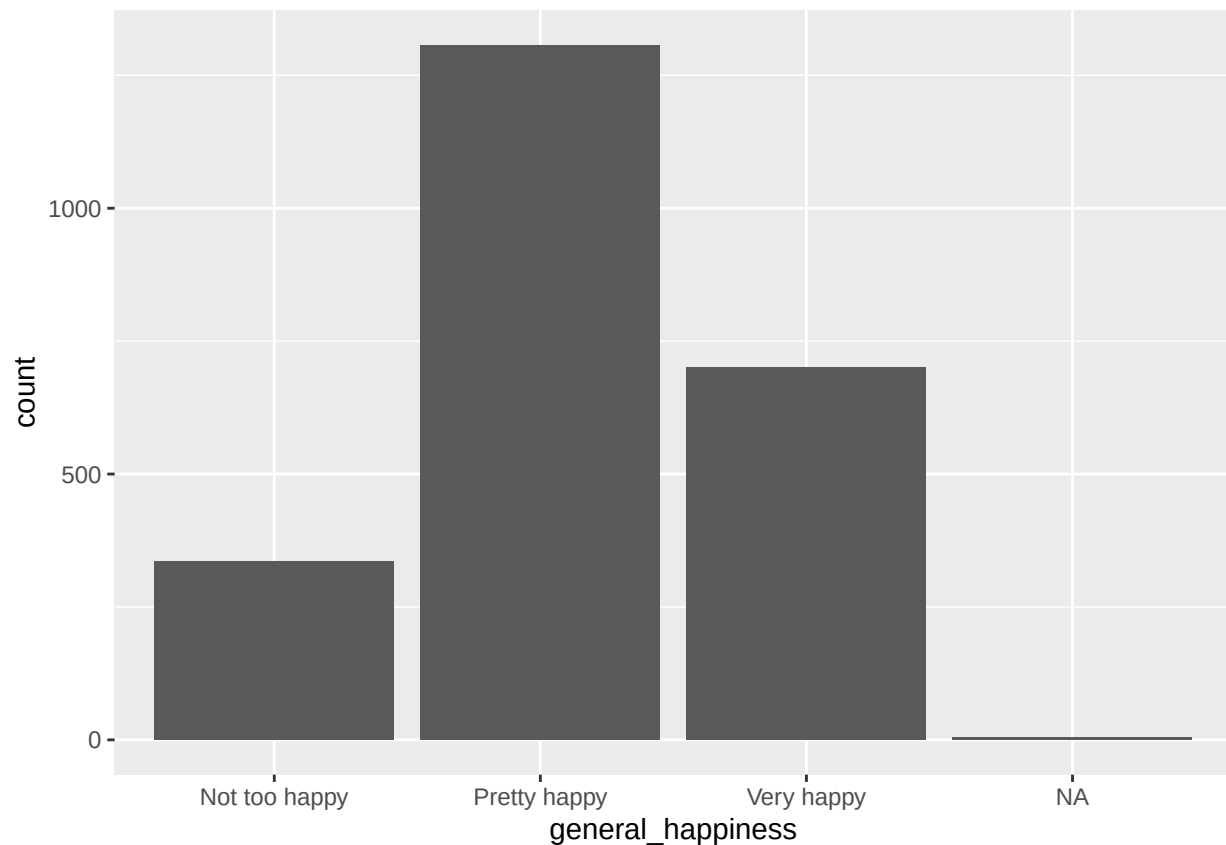
```
ggplot(data = GSS2, mapping = aes(x = marital_status, fill = marital_status)) +  
  geom_bar() +  
  scale_fill_viridis_d() +  
  labs(x = "Marital status", y = "Number of respondents",  
       title = "Distribution of marital status in the general social survey") +  
  coord_flip()
```

Distribution of marital status in the general social survey



Exercise: Make a bar graph of the variable `general_happiness`. You can have the bars go horizontally or vertically. If you discover that this variable has missing values, go ahead and drop the NAs. You can use the technique I described above the original NA drop section. Fill in the bars with color somehow. Add labels and a title.

```
# make the graph
ggplot(data = GSS, mapping = aes(x = general_happiness)) +
  geom_bar()
```

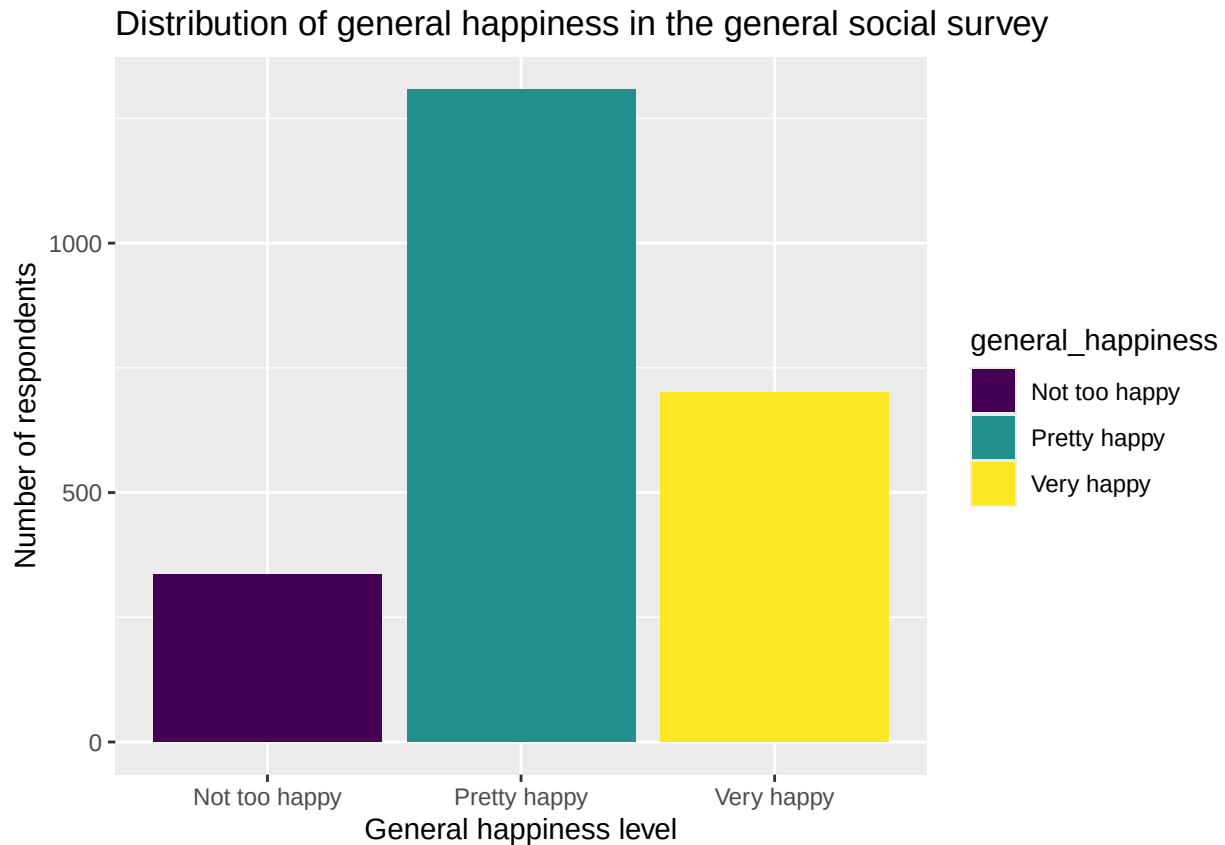


```
# drop missing vals
GSS3<- GSS %>%
  filter(general_happiness != "NA") %>%
  droplevels()

# verify dropped
GSS3 %>%
  distinct(general_happiness)

##   general_happiness
## 1      Pretty happy
## 2      Very happy
## 3      Not too happy

# add styles
ggplot(data = GSS3, mapping = aes(x = general_happiness, fill = general_happiness)) +
  geom_bar() + scale_fill_viridis_d() + labs(x = "General happiness level", y = "Number of respondents"
  title = "Distribution of general happiness in the general social survey")
```



Working towards re-ordering the bars and other improvements

So far, we have been using `geom_bar()` to create our bar plots. This geom is nice because it saves us from having to tell R to do any counting for us. In the example above, for instance, `geom_bar()` automatically counts the number of each survey respondents in each category for us and then uses those counts to create a bar plot.

There is another geom that can be used to create bar plots, called `geom_col()`. To use this geom, however, we first have to have R do some counting for us and then pipe the tallied data into the ggplot command. One might wonder why we might ever do this when we have `geom_bar()` to do the counting for us. The answer is that using `geom_col` gives us some additional flexibility to be able to add more features to our graphs. For example, below, we will learn how to reorder the bars in a bar plot, how to change from a “count” to a proportion, and how to add labels to the bars.

We get started by once again running this piece of code and noticing what it does.

```
GSS2 %>%
  summarize(n = n())
```

```
##           n
## 1 2346
```

What does this piece of code do?

The snippet of code counts the total number of rows in the new GSS2 dataframe and stores the count in a new column ‘n’.

We will add another line of code below to create a frequency table for `marital_status`.

```
GSS2 %>%
  group_by(marital_status)%>%
  summarize(n = n())
```

```
## # A tibble: 5 x 2
##   marital_status      n
##   <chr>          <int>
## 1 Divorced        403
## 2 Married         998
## 3 Never married   670
## 4 Separated        75
## 5 Widowed        200
```

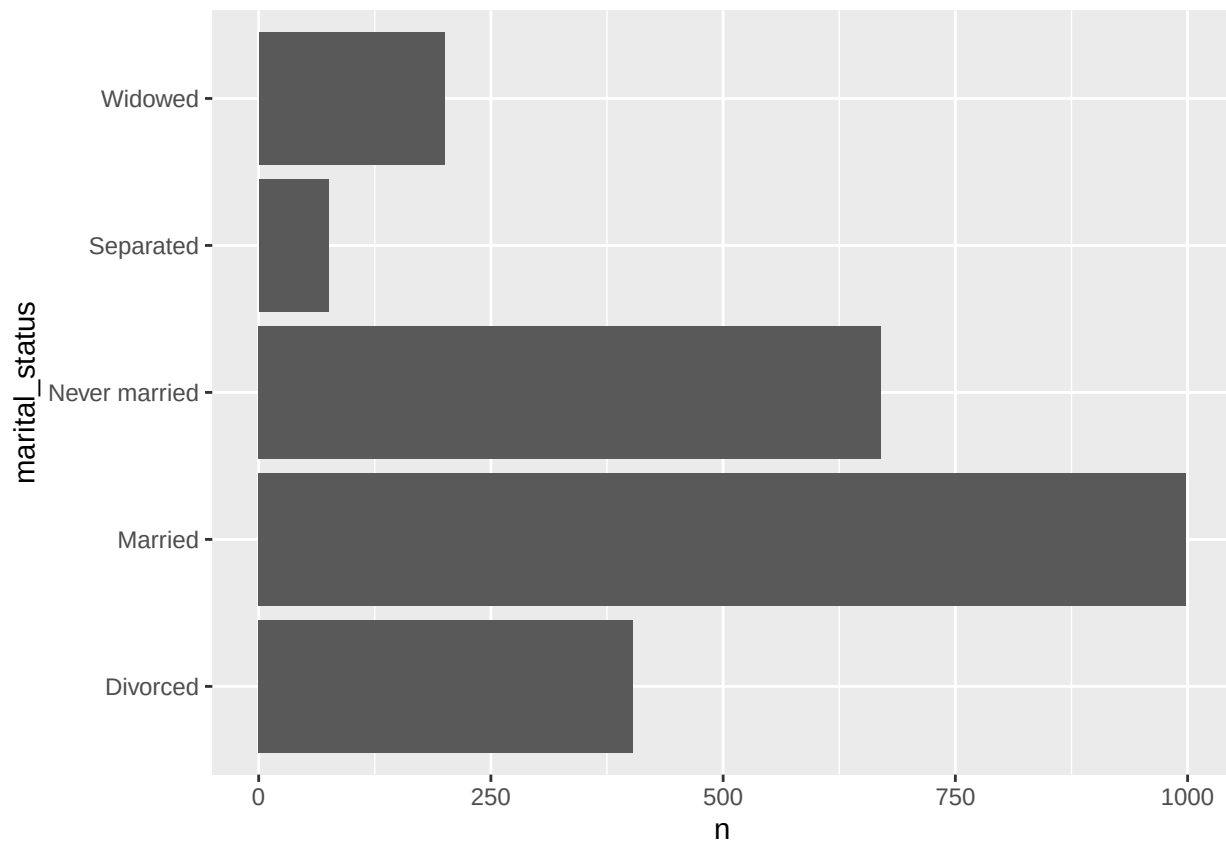
Next, we will add one more layer, `mutate(prop = n/sum(n))` to create a relative frequency table for `marital_status`.

```
GSS2 %>%
  group_by(marital_status)%>%
  summarize(n = n()) %>%
  mutate(prop = n/sum(n))
```

```
## # A tibble: 5 x 3
##   marital_status      n  prop
##   <chr>          <int> <dbl>
## 1 Divorced        403 0.172
## 2 Married         998 0.425
## 3 Never married   670 0.286
## 4 Separated        75 0.0320
## 5 Widowed        200 0.0853
```

Next, we will pipe our newly-created table into a `ggplot` function and replicate the graph we made at the end of the last section, now using `geom_col()` instead of `geom_bar`.

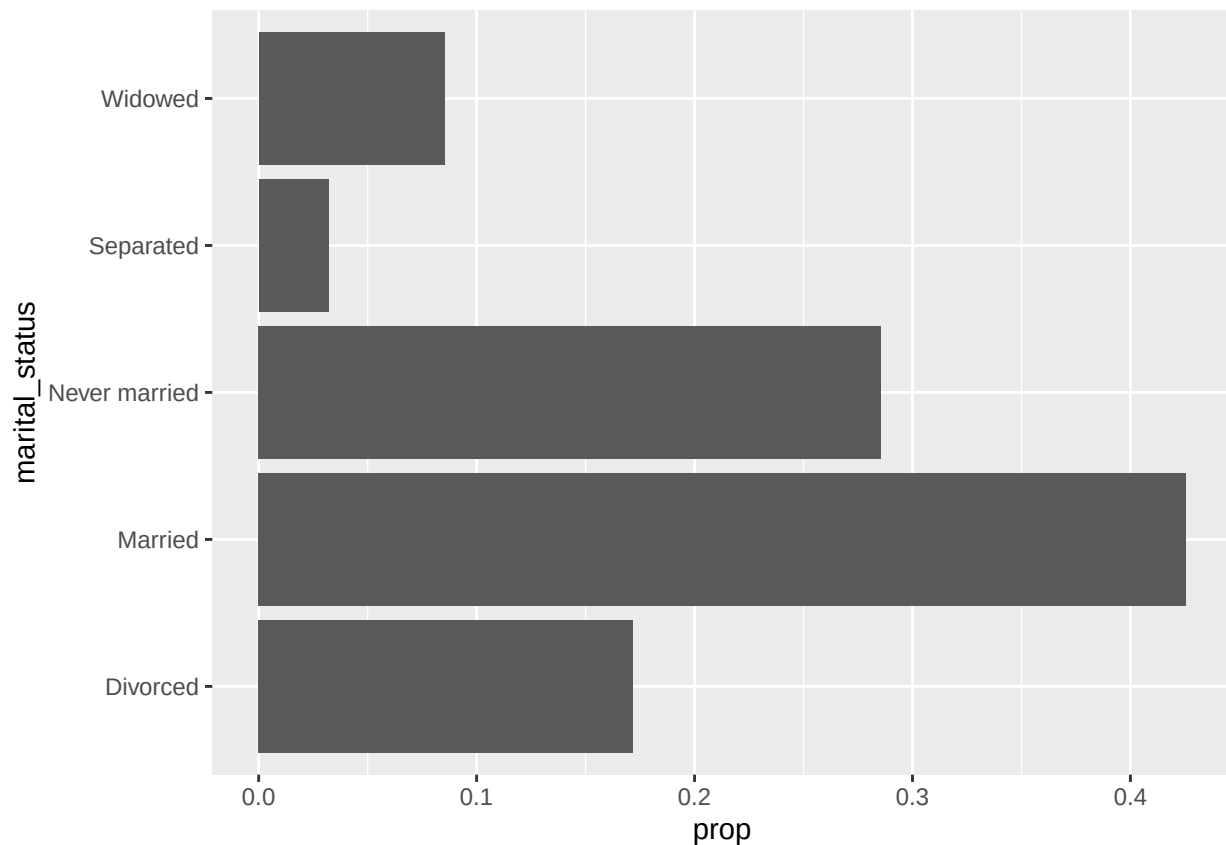
```
GSS2 %>%
  group_by(marital_status)%>%
  summarize(n = n()) %>%
  mutate(prop = n/sum(n)) %>%
  ggplot(mapping = aes(x = n, y = marital_status))+
  geom_col()
```

Notice that I have put the categories for `marital_status` on the y-axis to make the bars horizontal. What is going on the x-axis is the count in each category from the table that we made.

Below, copy and paste, and then modify the code from the previous chunk to have R graph the proportions of survey respondents in each category instead of the count of respondents in each category.

```
GSS2 %>%
  group_by(marital_status)%>%
  summarize(n = n()) %>%
  mutate(prop = n/sum(n)) %>%
  ggplot(mapping = aes(x = prop, y = marital_status))+
  geom_col()
```



Exercise: Make a frequency table and then a relative frequency table for the variable `general_happiness`. Then make a bar graph for that variable using `geom_col()`.

```
# frequency table
GSS2 %>%
  group_by(general_happiness)%>%
  summarize(n = n())

## # A tibble: 4 x 2
##   general_happiness    n
##   <chr>             <int>
## 1 Not too happy      336
## 2 Pretty happy     1306
## 3 Very happy        700
## 4 <NA>              4

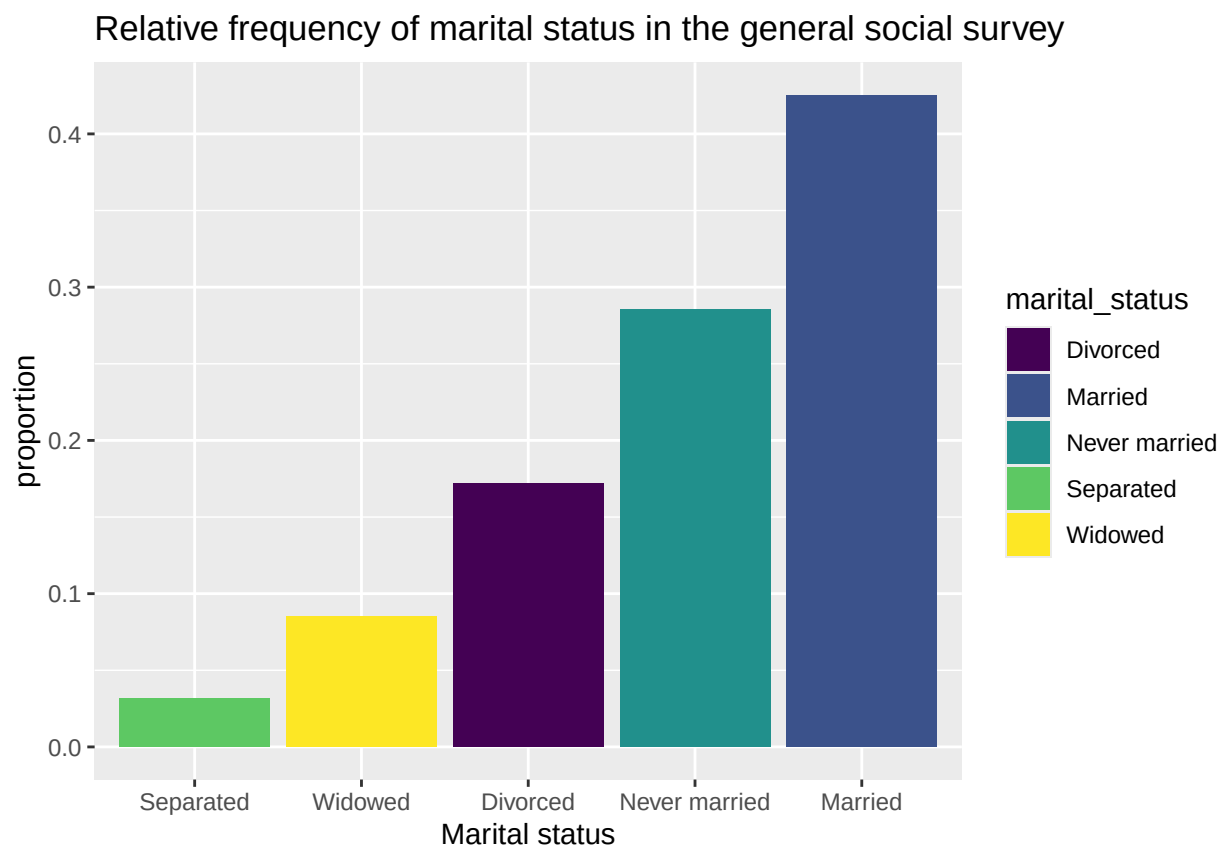
#relative frequency table
GSS2 %>%
  group_by(marital_status) %>%
  summarize(n = n()) %>%
  mutate(prop = n / sum(n)) # relative frequency

## # A tibble: 5 x 3
##   marital_status    n  prop
##   <chr>          <int> <dbl>
## 1 Divorced       403 0.172
## 2 Married       998 0.425
## 3 Never married  670 0.286
```

```
## 4 Separated          75 0.0320
## 5 Widowed           200 0.0853

# bar graph for (relative frequency)

GSS2 %>%
  group_by(marital_status) %>%
  summarize(n = n()) %>%
  mutate(prop = n / sum(n)) %>%
  ggplot(mapping = aes(x = fct_reorder(marital_status, prop), y = prop, fill = marital_status)) +
  geom_col() +
  scale_fill_viridis_d() +
  labs(
    title = "Relative frequency of marital status in the general social survey",
    x = "Marital status", y = "proportion")
```

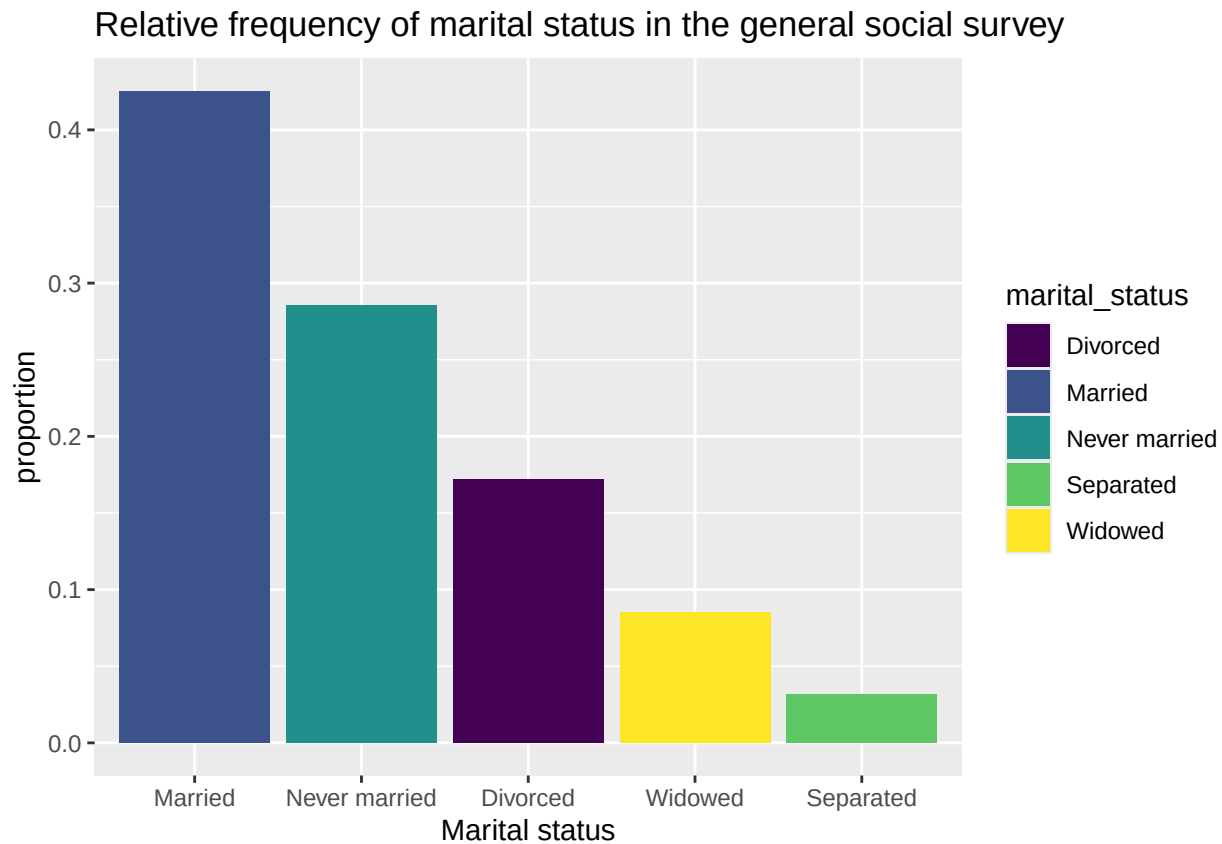


Reordering the bars of a bar plot

Sometimes, we might want to reorder the bars in a bar graph to make the visualization easier to read and interpret. Below, copy and paste the code from your last graph of `marital_status`. Then modify it so that instead of reading `y = marital_status`, you have `y = fct_reorder(marital_status, prop)` instead. This piece of code will reorder the bars of `marital_status` by the proportions in our table, in descending order from top to bottom.

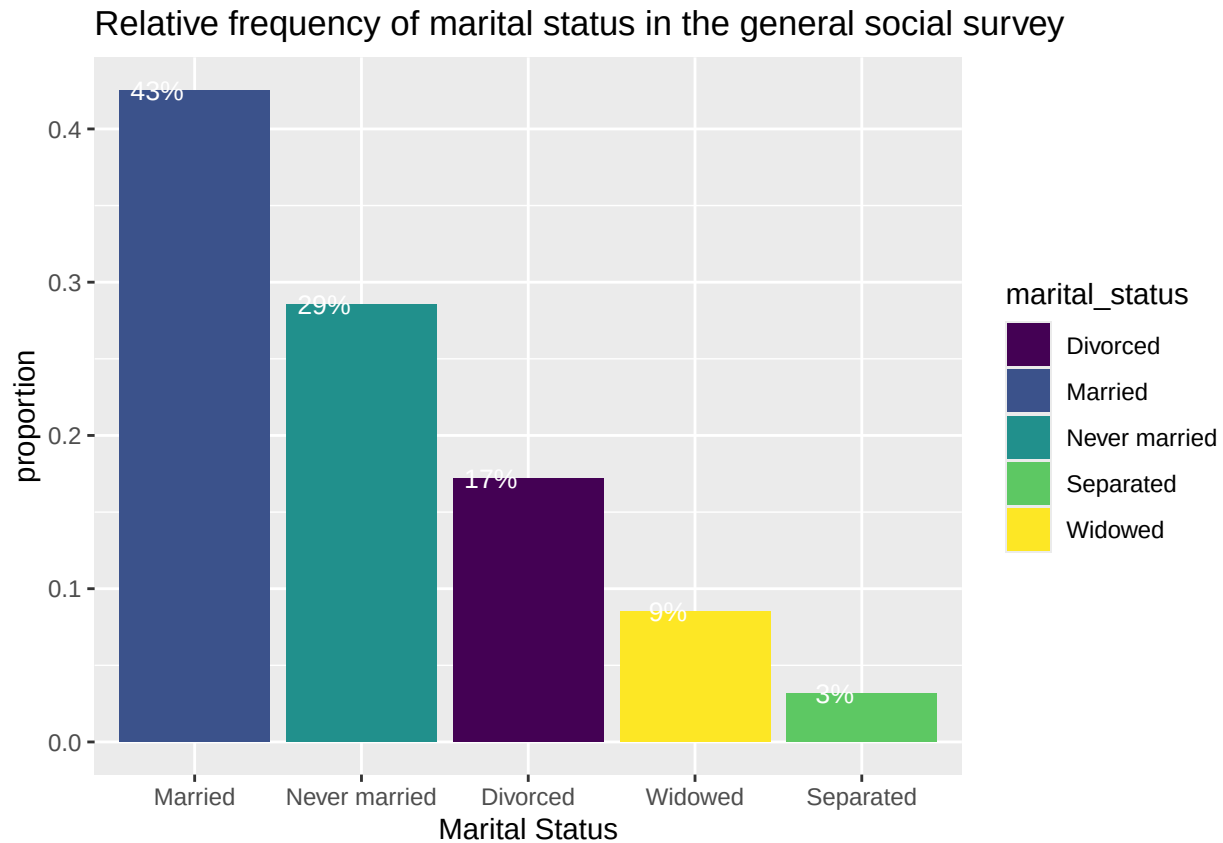
```
GSS2 %>%
  group_by(marital_status) %>%
  summarize(n = n()) %>%
  mutate(prop = n / sum(n)) %>%
```

```
ggplot(mapping = aes(x = fct_reorder(marital_status, desc(prop)), y = prop, fill = marital_status)) +
  geom_col() +
  scale_fill_viridis_d() +
  labs(
    title = "Relative frequency of marital status in the general social survey",
    x = "Marital status", y = "proportion")
```



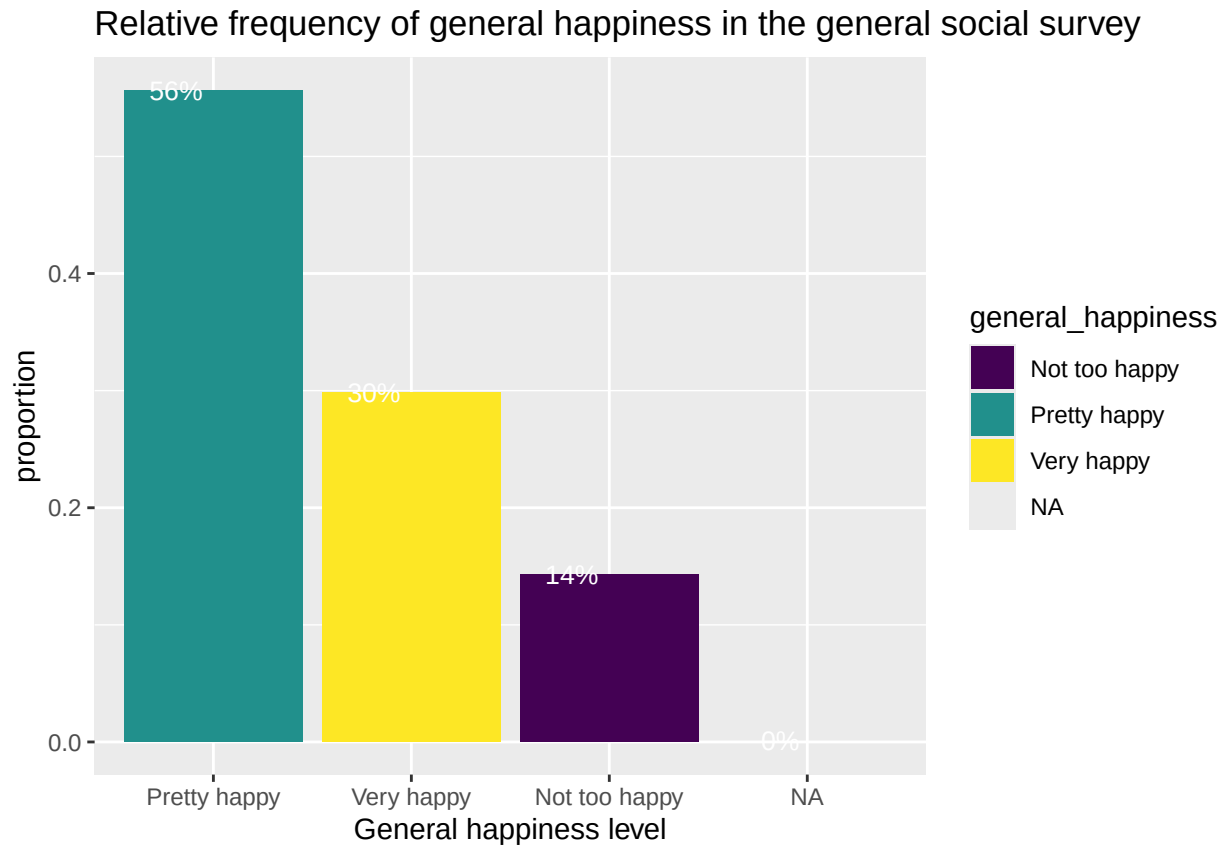
Finally, paste the code from the above chunk below, and add one more layer at the end. This is the line of code for the extra layer: `geom_text(aes(label = paste(round(prop*100), "%", sep = "")), hjust = 1.2, color = "white", size = 3.5)`, without the backticks, to see what happens.

```
GSS2 %>%
  group_by(marital_status) %>%
  summarize(n = n()) %>%
  mutate(prop = n / sum(n)) %>%
  ggplot(mapping = aes(x = fct_reorder(marital_status, desc(prop)), y = prop, fill = marital_status)) +
  geom_col() +
  scale_fill_viridis_d() +
  labs(
    title = "Relative frequency of marital status in the general social survey",
    x = "Marital Status", y = "proportion") +
  geom_text(aes(label = paste(round(prop * 100), "%", sep = "")),
    hjust = 1.2, color = "white", size = 3.5)
```



Exercise: Practice reordering the bars of your `general_happiness` variable. Add the extra layer indicating the percentage represented by each bar.

```
GSS2 %>%
  group_by(general_happiness) %>%
  summarize(n = n()) %>%
  mutate(prop = n / sum(n)) %>%
  ggplot(mapping = aes(x = fct_reorder(general_happiness, desc(prop)), y = prop, fill = general_happiness)) +
  geom_col() +
  scale_fill_viridis_d() +
  labs(
    title = "Relative frequency of general happiness in the general social survey",
    x = "General happiness level", y = "proportion") +
  geom_text(aes(label = paste(round(prop * 100), "%", sep = "")),
    hjust = 1.2, color = "white", size = 3.5)
```



ONE LAST THING! Sometimes, instead of using a bar graph for visualizing a categorical variable, we can use a graph called a *Cleveland dotplot*. Cleveland dotplots are especially useful for when there are a large number of categories. Practice making a Cleveland dotplot below. The code is exactly the same as for our bar graphs (the `geom_col` method) but instead of using `geom_col()`, just use `geom_point()`. Copy and paste your code from the previous chunk and change `geom_col()` to `geom_point()` to see what happens. You will notice that some of the line(s) of code are now unnecessary. Delete the unnecessary line(s) of code.

```
GSS2 %>%
  group_by(marital_status) %>%
  summarize(n = n()) %>%
  mutate(prop = n / sum(n)) %>%
  ggplot(mapping = aes(x = prop, y = fct_reorder(marital_status, prop))) +
  geom_point(size = 4, color = "blue") + # dot characteristics
  labs(title = "cleveland dp of marital status in the GSS", x = "proportion", y = "marital status")
```

cleveland dp of marital status in the GSS

